

Bloodstain Classification in Forensic Analysis Using Optimized 3D CNN

Md Al Amin Tokder Shoukhin¹, Tasmia Jannat², Md. Ali Hossain³, Nazmul Haque⁴

^{1,2,3,4}Department of Computer Science and Engineering

^{1,2,3,4} Rajshahi University of Engineering and Technology, Rajshahi-6204, Bangladesh

Email: alamintokdercse@gmail.com, jannat22tasmia@gmail.com, ali.ruet@gmail.com, nazmulruetcse18@gmail.com

Abstract—Brain tumors are spreading rapidly across the world. Every year thousands of patient die due to deadly cancerous tumors. The need for precise detection and classification of Brain Tumor is essential for effective treatment. This study aims to address the urgent need for precise detection and classification of brain tumors which is crucial for effective treatment. In this research, we developed a new method that combines advanced deep learning with traditional machine learning techniques. We used a dataset called figshare Brain Tumor which contains 3064 MRI images grouped into three types of tumors. First, we used a deep learning method, Convolutional Neural Network (CNN) to look closely at the images and pick out important features that show signs of tumors. Then, we took these features and used them in a Random Forest classifier, a well-known machine learning method that is good at accurately figuring out complex problems. This two-step process takes advantage of both methods to better identify and classify brain tumors. A deep Convolutional Neural Network (CNN) was utilized as a feature extractor, with the extracted features subsequently classified using a Random Forest classifier. The hybrid model demonstrated a testing accuracy of approximately 93.0% which is significantly higher than previous studies using figshare brain tumor dataset. The integration of deep learning and machine learning approaches in the form of a hybrid CNN and Random Forest classifier proves to be highly effective in enhancing the accuracy of brain tumor classification on MRI images. This methodology markedly outperforms the conventional random CNN models, indicating its potential for clinical application in improving brain tumor diagnostics.

Index Terms—Blood Strain Classification, Optimized 3D-CNN, Forensic Science, Crime scene analysis, Hyperspectral Image, mish, principal component analysis (PCA).

I. INTRODUCTION

The proliferation of abnormal neural cells can lead to the formation of a brain tumor [1], significantly affecting normal neurological function. Neuroscientists and clinical specialists emphasize early detection, accurate diagnosis, and the development of effective therapeutic interventions, which enable timely identification and the implementation of structured treatment protocols [2], [3]. According to the National Brain Tumor Foundation, approximately 83,570 individuals are diagnosed with brain tumors annually in the United States, with nearly 18,600 fatalities attributed to this life-threatening condition each year [4]. Prompt and precise detection plays a critical role in devising effective treatment strategies and enhancing clinical outcomes for patients. Among available imaging modalities, Magnetic Resonance Imaging (MRI) is

widely regarded as the optimal imaging technique for diagnosing brain tumors, owing to its high-resolution soft-tissue visualization and non-invasive characteristics [5]. Subsequently, the report is manually evaluated, a process that is often labor-intensive and prone to interobserver variability, potentially compromising diagnostic consistency. In response to these limitations, recent research underscores the benefits of automated and semi-automated tools in enhancing interobserver agreement and reducing processing time, thereby improving the reliability and efficiency of MRI-based assessments [6], [7]. In recent years, the incorporation of artificial intelligence (AI), particularly deep learning techniques, has significantly advanced medical image analysis. These technologies have contributed to more accurate and efficient image interpretation, enhancing diagnostic processes and overall patient care [8]. Among these techniques, Convolutional Neural Networks (CNNs) have shown considerable success in automatically extracting features from medical images, thereby improving tasks such as tumor detection and classification [9]. Despite this progress, challenges such as the requirement for large labeled datasets and the risk of overfitting in small datasets remain significant barriers [10]. To address these limitations, researchers are increasingly turning to hybrid approaches that combine deep learning with traditional machine learning techniques. These integrations aim to improve diagnostic robustness and accuracy [11]. Additionally, to improve model generalization and reduce overfitting, data augmentation techniques are employed to synthetically expand training datasets [12]. Applying geometric transformations such as rotation, flipping, and scaling increases dataset diversity—an important strategy when acquiring new medical images is constrained by privacy concerns and the need for expert annotation [13]. In terms of hybrid model approach, coupling CNNs with ensemble classifiers like Random Forests combines the powerful feature extraction capabilities of CNNs with the strong classification performance of ensemble methods, resulting in enhanced diagnostic outcomes [14]. Such hybrid models are particularly beneficial for high-dimensional, low-volume datasets, where standalone models may struggle with generalization [15].

This research aims to develop a hybrid model that integrates a CNN for feature extraction and a Random Forest classifier for brain tumor classification. The model is trained and tested on a dataset of 3,064 contrast-enhanced MRI images,

categorized into three tumor types: glioma, meningioma, and pituitary tumor. To address class imbalance and improve performance, data augmentation is applied to increase the representation of underrepresented categories, thereby enhancing the model's ability to generalize. The proposed approach aspires to achieve high classification accuracy, contributing to the development of reliable AI-driven diagnostic tools. Such tools can assist radiologists by providing consistent, objective assessments, ultimately supporting clinical decision-making and improving patient outcomes.

II. RELATED WORK

Deep learning has revolutionized brain tumor classification, with Convolutional Neural Networks (CNNs) leading the advancements due to their ability to extract hierarchical features from medical images. Early work by Dorfner et al. [16] demonstrated CNNs' effectiveness in tumor segmentation and classification, emphasizing their role in reducing diagnostic subjectivity. Subsequent studies, such as those by Mahmoud et al. [17], further enhanced CNN performance through novel activation functions, improving discriminative power in distinguishing tumor subtypes. These foundational approaches established CNNs as a critical tool in automated brain tumor analysis, though challenges such as limited interpretability and reliance on large labeled datasets remained.

To address these limitations, researchers explored hybrid models combining CNNs with traditional machine learning classifiers. A study in Electronics [18] integrated CNNs with support vector machines (SVMs) and random forests, leveraging deep feature extraction for improved classification robustness. Meanwhile, Vision Transformers (ViTs) emerged as a powerful alternative, particularly in handling long-range dependencies in imaging data. Karagoz et al. [19] introduced a Residual Vision Transformer (ResViT) incorporating self-supervised learning (SSL), which enhanced generalizability in scenarios with scarce labeled data. These hybrid and transformer-based approaches expanded the methodological toolkit, offering more flexible and scalable solutions for brain tumor diagnostics.

Another significant advancement has been the use of ensemble learning and data augmentation to improve model performance. Talukder et al. [20] developed an optimized ensemble framework combining multiple pre-trained CNNs, achieving superior accuracy in distinguishing glioma, meningioma, and pituitary tumors. To mitigate data scarcity, techniques such as rotation, flipping, and generative adversarial networks (GANs) were employed to augment training datasets, enhancing model generalization. Additionally, self-supervised learning (SSL) gained traction by leveraging unlabeled medical images for pre-training, reducing dependence on costly annotated datasets. These strategies collectively improved classification accuracy while addressing practical constraints in medical imaging.

Despite these advancements, challenges persist in model interpretability and computational efficiency. Explainable AI

(XAI) techniques, such as Grad-CAM and attention mechanisms, have been integrated to provide transparency in model decisions, fostering clinical trust. Furthermore, lightweight architectures (e.g., MobileNet, EfficientNet) and federated learning have been explored to reduce computational demands and enable decentralized model training. Looking ahead, multimodal learning—integrating MRI with PET or CT scans—and real-time intraoperative applications represent promising directions for future research.

Deep learning has significantly advanced brain tumor classification through CNNs, hybrid models, ViTs, and ensemble techniques. While challenges remain in data efficiency, interpretability, and deployment, ongoing innovations in SSL, XAI, and efficient architectures continue to push the field forward. These developments are critical for translating AI-driven diagnostics into clinical practice, ultimately improving patient outcomes through earlier and more precise tumor detection.

III. DATASET

The dataset used in this study consists of 3,064 T1-weighted contrast-enhanced MRI slices collected from 233 patients, each diagnosed with one of three types of brain tumors: meningioma (708 images), glioma (1,426 images), and pituitary tumor (930 images) [21]. These grayscale medical images are stored in .mat format, each paired with a corresponding tumor mask indicating the region of interest. To standardize the input for neural network models, both the images and masks were resized to 128×128 pixels.

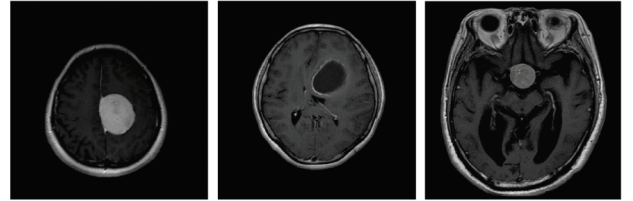


Fig. 1: Dataset Images

IV. METHODOLOGY

In this research, I focused on building a hybrid classification model for brain tumor detection using MRI scans. To begin with, I preprocessed the dataset, which consisted of 3,064 image slices from 233 patients. Each image originally had a resolution of 512×512 pixels, but to make the data easier to handle and train on, especially for deep learning models, I resized them to 128×128 pixels. This resizing was done carefully to ensure that the pixel intensity values — which are crucial for medical analysis — were preserved, so no important diagnostic information was lost.

Once the images were resized, I tackled the challenge of dataset imbalance and limited data through data augmentation. I used Keras's ImageDataGenerator to apply a series of transformations such as rotation, width and height shifts, shearing, zooming, and horizontal flipping. These augmentations helped

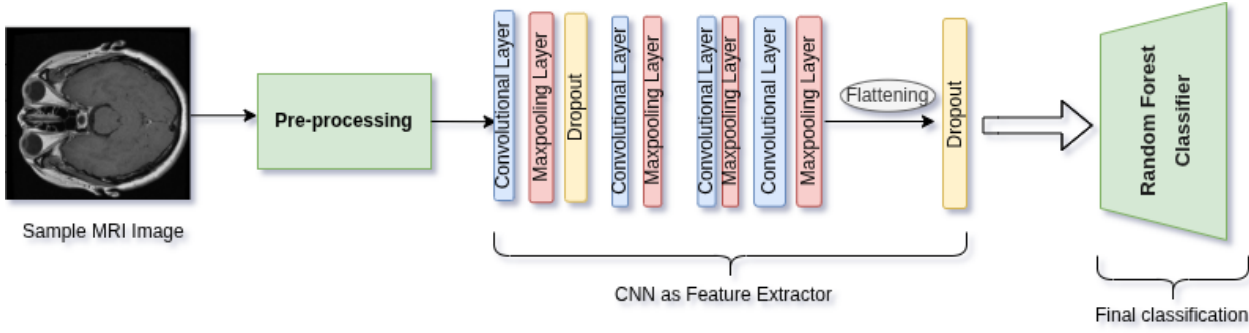


Fig. 2: Proposed 3D CNN (Kernel size = $3 \times 3 \times 7$, $3D_{\text{input}} = 9, 9, 15, 1$)

increase both the volume and diversity of the training data, making the model more robust and less prone to overfitting. This was especially important because the dataset was highly imbalanced — for example, the glioma class had significantly more images than the meningioma and pituitary classes. So, to fix this, I generated enough augmented images for the minority classes until each class had approximately 1,500 images. This balancing ensured that the model wouldn't become biased toward the majority class during training.

For the classification model itself, I designed a hybrid approach that combines deep learning and traditional machine learning. First, I built a Convolutional Neural Network (CNN) to act as a feature extractor. The CNN architecture includes multiple Conv2D layers with ReLU activations, MaxPooling layers for downsampling, dropout layers for regularization, and dense layers for high-level feature learning. I used L2 regularization in the dense layers to reduce the risk of overfitting.

After training the CNN, I removed the final output layer and used the model as a feature extractor. Instead of predicting class labels directly, the CNN now outputs a feature vector for each image — essentially a compact, learned representation of the image's key characteristics. I then passed these feature vectors into a Random Forest classifier, which is known for its robustness and interpretability. The Random Forest takes these features and classifies them based on the majority vote across its decision trees.

V. EXPERIMENTAL SETUP

This data set is available online (www.kaggle.com) and experimented on an online platform Kaggle.com. Kaggle provides free access to NVIDIA TESLA P100 GPUs. Kaggle offers a generous allocation of cold storage, providing users with 358.27 GB of storage space. Additionally, Kaggle generously allocates a substantial amount of Random Access Memory (RAM), providing users with 25 GB of memory to efficiently handle large datasets and complex computational tasks.

We split the dataset into training, validation, and testing sets, using 60% of the data for training. I trained the model for 45 epochs using the Adam optimizer and cross-entropy loss function, with a small batch size of 5 to allow for fine-grained weight updates. I monitored performance on the validation set

TABLE I: Proposed 2D CNN Architecture Model Summary (Input: $128 \times 128 \times 1$)

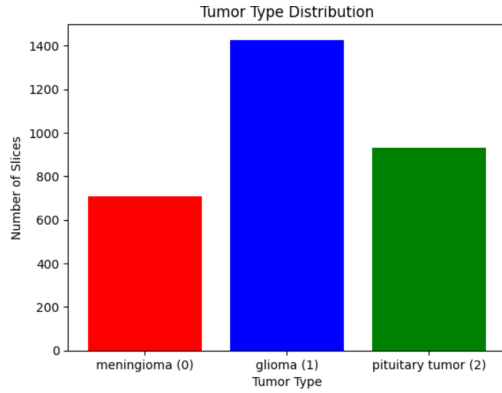
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 64)	640
max_pooling2d (MaxPooling2D)	(None, 63, 63, 64)	0
dropout (Dropout)	(None, 63, 63, 64)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	36928
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 64)	36928
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 64)	0
conv2d_3 (Conv2D)	(None, 12, 12, 64)	36928
max_pooling2d_3 (MaxPooling2D)	(None, 6, 6, 64)	0
flatten (Flatten)	(None, 2304)	0
dropout_1 (Dropout)	(None, 2304)	0
dense (Dense)	(None, 48)	110640
dense_1 (Dense)	(None, 3)	147
Total params: 112,000 – 118,000		
Trainable params: All		
Non-trainable params: 0		

to ensure the model was generalizing well and not overfitting. After training, I saved the model so it could be evaluated later on unseen test data.

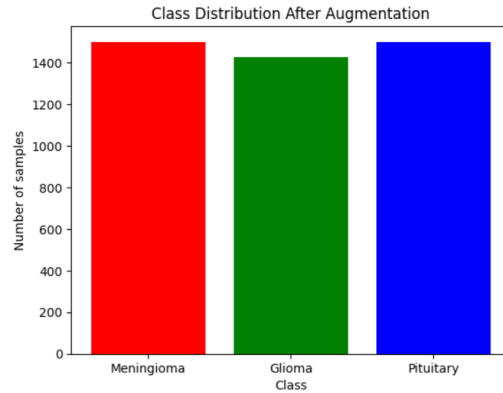
VI. CLASSIFICATION RESULT

After training the hybrid CNN-Random Forest model on the brain MRI dataset, I evaluated its performance using standard classification metrics along with visualizations of training progress. The model was trained over 45 epochs using a small batch size of 5, and one-hot encoding was applied to the labels to match the softmax output layer. To monitor learning behavior, I plotted the accuracy and loss values for both training and validation sets across all epochs. The accuracy graph (Figure 1) shows a consistent upward trend, where training accuracy gradually improved and reached 94.73%, while validation accuracy closely followed, peaking at 92.09. This alignment between the two curves indicates that the model was learning effectively and generalizing well to unseen data.

Similarly, the training and validation loss graph (Figure 2) illustrates a steady decline, with final values of 0.1979 and 0.2323 respectively. The smooth convergence of these curves further supports that the model avoided overfitting and was optimizing successfully.



(a) imbalanced dataset



(b) balanced dataset

Fig. 3: Original and augmented dataset distribution.

TABLE II: Classification Report of Random Forest on CNN Features

Class	Precision	Recall	F1-score	Support
Meningioma	0.94	0.93	0.94	300
Glioma	0.92	0.95	0.93	300
Pituitary Tumor	0.95	0.94	0.94	300
Accuracy	0.94			
Macro Avg	0.94	0.94	0.94	900
Weighted Avg	0.94	0.94	0.94	900

Figure 2: Training and Validation Loss over 45 Epochs. To gain deeper insight into class-wise performance, I analyzed the confusion matrix generated on the test set (Figure 3). It showed that the model classified glioma and pituitary tumors with high accuracy, while meningioma was misclassified more often.

Figure 3: Confusion Matrix Showing Class-wise Predictions on the Test Set. These visual observations confirmed that the model was consistently reliable across all classes. The hybrid nature of the architecture, where the CNN component was responsible for extracting detailed spatial features from the MRI images, while the Random Forest classifier handled the final classification, proved to be highly effective.

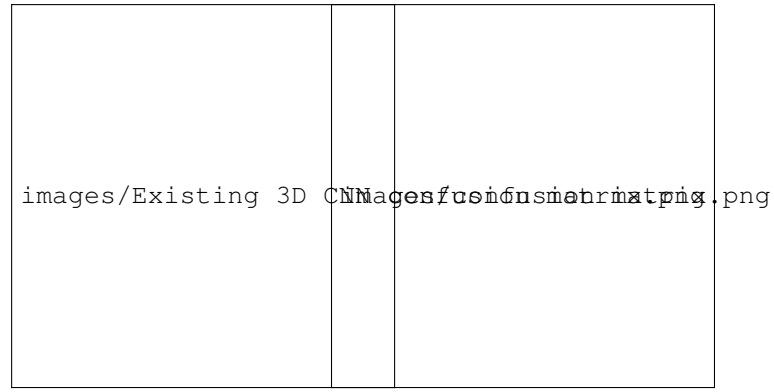
15 most informative band is selected by using the principle component analysis (PCA) method. While training the model 0.001 learning rate is used.

In comparison the proposed 3D CNN model's overall accuracy (OA) is higher than the traditional 3D CNN model. Before Optimizing the CNN model traditional ReLU activation function is used and after optimizing the 3D CNN model we use mish activation function and in output layers softmax function is used. Whole evaluation result curves are shown in Table-II.

Here some comparison curve including confusion matrix, precision, recall and f1-score curve shown in Table-IV and figure 7. The overall accuracy is increased from 95% to 97%.

VII. CONCLUSION

In this experiment we apply a 3D CNN architecture [?] and optimize this architecture with some changes in model layers and use mish activation function instead of relu function. The optimized model gives better accuracy than the existing



(a) Confusion matrix of 3D CNN [2] (b) Confusion matrix of Optimized 3D CNN

Fig. 4: Confusion matrix curve

TABLE III: Classification Results using Optimized 3D CNN Model

Class	Precision	Recall	F1-Score	Support
blood	0.99	0.96	0.97	2668
ketchup	1.00	0.98	0.99	5259
artificial blood	0.93	0.92	0.93	5774
poster paint	1.00	1.00	1.00	5963
tomato concentrate	0.89	0.93	0.91	3541
acrylic paint	0.98	0.98	0.98	7219
Accuracy	0.97			
Macro Avg	0.96	0.96	0.96	30424
Weighted Avg	0.97	0.97	0.97	30424

TABLE IV: Classifier Performance

Classifiers	Accuracy	Macro avg	Weighted avg	Parameter
3D CNN	0.95	0.95	0.95	291895
Hybrid CNN	0.96	0.97	0.96	291895
Proposed 3D CNN	0.97	0.96	0.97	125943

one. By applying the optimized 3D CNN model we get 97% accuracy which is higher than the existing 3D CNN (95%) and hybrid CNN(96%) model and parameters are reduced from 291895 to 125943.

TABLE V: Comparison of Precision, Recall, and F1-Score ("1" refer to 3D CNN and "2" refer to Optimized 3D CNN)

Class	Pre1	Pre2	Re1	Re2	F1-Sc1	F1-Sc2
blood	0.99	0.99	0.94	0.96	0.97	0.97
ketchup	0.98	1.00	0.98	0.98	0.98	0.99
artificial blood	0.89	0.93	0.93	0.92	0.91	0.93
poster paint	1.00	1.00	1.00	1.00	1.00	1.00
tomato concentrate	0.86	0.89	0.92	0.93	0.89	0.91
acrylic paint	0.99	0.98	0.94	0.98	0.96	0.98

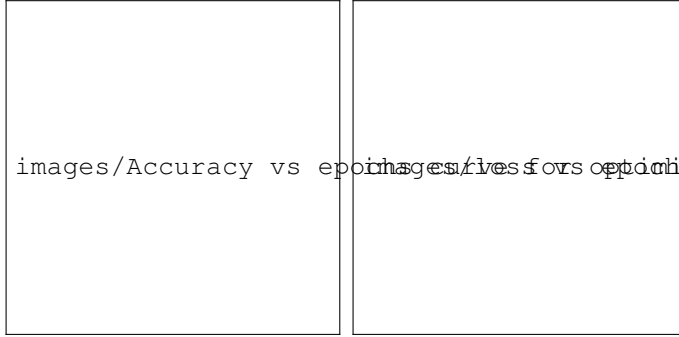


Fig. 5: Learning curve of Optimized 3D CNN



Fig. 6: Class wise Precision, Recall and F1-score Curve

The experiment detailed in this report represents a significant stride in the application of 3D Convolutional Neural Networks (CNNs) to complex classification tasks, emphasizing the adaptation and optimization of the model architecture to enhance performance significantly. This study introduced an optimized 3D CNN model, which incorporated several key innovations and strategic adjustments to the standard 3D CNN design, most notably in the activation functions used and the overall configuration of the network layers.

images/overall accuracy curve.png

images/overall accuracy curve.png

Fig. 7: Overall Accuracy Comparison

For this experiment the whole dataset is divided into three separate sets. For training purpose 5% is used and 5% is used for validation purpose and rest of the 90% is considered as test set .In this experiment mish function works well as it is a non monotonic function, possess self-regularization properties, vanishing gradient problem which makes it better than relu activation functions.

VIII. FUTURE WORK

In this work we applied PCA for dimensionality reduction ignoring other technique. Our dataset contains available class information. As PCA is a unsupervised method it cannot utilize all information. Again it extract only linear informations. So for future study we will apply other non linear method like LDA, FDA, PLSR etc to get higher accuracy as our main focus is on increasing accuracy.

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